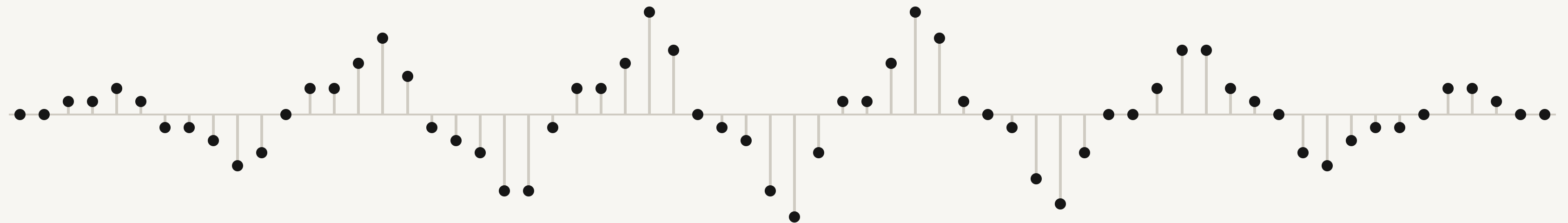


Demystifying Game Audio Codecs

A Practical Guide

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About me

I've worked on game audio in commercial engines and proprietary ones.

Riot Games	Unreal Audio, Wwise Unannounced R&D
High Moon Studios	Proprietary engine tech Call of Duty: Modern Warfare 4, Black Ops 6 and Black Ops 7
Gas Powered Games	Unity and Wwise Unpublished title

What a codec setting affects

Most projects pick a codec once, from a dropdown. That one setting decides:

File size	Install footprint and patch size
Streaming	Disk and I/O bandwidth under load
Memory	RAM held by loaded sounds
CPU	Decode cost for every voice that is playing
Platform compliance	Format requirements and certification
Sound quality	What players actually hear

Outline

Four sections. No code, and not too deep on the technical side.

01	The basics	What audio data is, and the problems codecs are meant to solve
02	The past	Codecs in games since sample playback took over, and what persists today
03	The modern toolbox	What teams can use now, and strategies for adopting it
04	The frontier	What is on the horizon, and what to keep an eye on

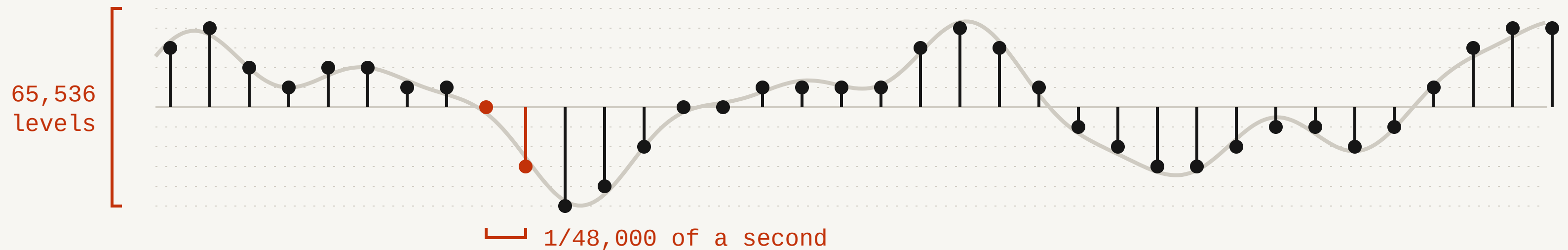
01

The basics

Raw PCM, how codecs work and the problems they solve

What is digital audio?

Pulse-code modulation (PCM). The amplitude of the analog signal is sampled at uniform intervals, and each sample is quantized to the nearest value within a range of digital steps.



Sample rate

How often the wave is measured:
48,000 times a second in most games.

Bit depth

How precisely each measurement is
stored. 16 bits gives 65,536 levels.

Channels

How many waveforms play at once.
Stereo is 2, and 5.1 is 6.

The cost of PCM: memory

48,000

×

16

×

2

=

1,536 kbps

samples per second

bits per sample

channels

uncompressed stereo, or 192 KB every second

Content	Size as PCM
One minute, stereo	11.5 MB
One hour, stereo	691 MB
One hour, 5.1	2.07 GB

For scale: all the sound RAM a PlayStation 2 game had



That RAM would hold about 11 seconds of this audio. Then multiply by every variation, language and platform.

The cost of PCM: streaming

Streaming means two things in games: reading audio from disk as it plays, and sending it over a network. Raw PCM makes both harder.

From disk

192 KB/s

per stereo stream, read constantly

- Competes for I/O with textures and level loads
- Hard drives and optical discs pay for every seek
- Each stream holds a RAM buffer: 2 s is 384 KB
- Nothing shrinks on disk: Titanfall shipped 35 GB

Over the network

768 kbps

per mono voice, sent constantly

- Upstream bandwidth is shared with game traffic
- A 20 ms frame (1,920 bytes) overflows one packet
- No built-in recovery: a lost packet is a gap
- Fixed bitrate: it cannot adapt to a bad connection

From disk, PCM buys you zero decode CPU. Over the internet, that is rarely worth the bandwidth.

A tradeoff in the wild: Titanfall

48 GB

35 GB

≈ 73%

PC install size, 2014

of it was uncompressed audio

of the install

The choice	Ship the PC audio uncompressed
The reason	Decoding costs CPU, and the minimum-spec PC had two cores
The trade	Disk space and download size, spent to keep CPU free for the game

“On a two-core [machine] with where our min spec is, we couldn't dedicate those resources to audio.”

Richard Baker, lead engineer at Respawn, to Eurogamer. Via The Escapist, March 2014.

Codec basics: types of codec

Every codec is either lossless or lossy, and lossy codecs come in two kinds.

Lossless	Every sample comes back exactly
Lossy: predictive	Predicts each sample, then stores a rough correction
Lossy: perceptual	Drops what the ear won't notice

Codec basics: compression ratios

The original size against the compressed size. One minute of stereo audio at 48 kHz, 16-bit:

PCM, uncompressed	1 : 1		11.5 MB
FLAC	~2 : 1		~5.8 MB
ADPCM	4 : 1		2.9 MB
Vorbis or Opus, 128 kbps	12 : 1		0.96 MB
Opus, 64 kbps	24 : 1		0.48 MB

ADPCM's ratio is fixed. Lossless depends on the material. Perceptual ratios are a setting: halve the bitrate, double the ratio.

Codec basics: quality and bitrate

Bitrate is how many bits each second of sound gets. More bits sound better, until you can't hear the difference.

Typical bitrates	
Opus: speech	28–40 kbps
Opus: mono music	48–64 kbps
Opus: stereo music	64–128 kbps
ADPCM: stereo, fixed	384 kbps
PCM: stereo	1,536 kbps

Constant or variable

Constant spends the same bits every second, so size is predictable. Variable spends more on busy passages and less on quiet ones. A quality slider means variable.

Transparent

The point where listeners can't tell it from the original. Past it, more bits only cost size.

Too low

Smearred transients, swirly highs, a dull top end.

Opus ranges: [IETF RFC 7587](#), at 20 ms frames, full bandwidth. Xiph calls Opus at 128 kbps VBR close to transparent: wiki.xiph.org

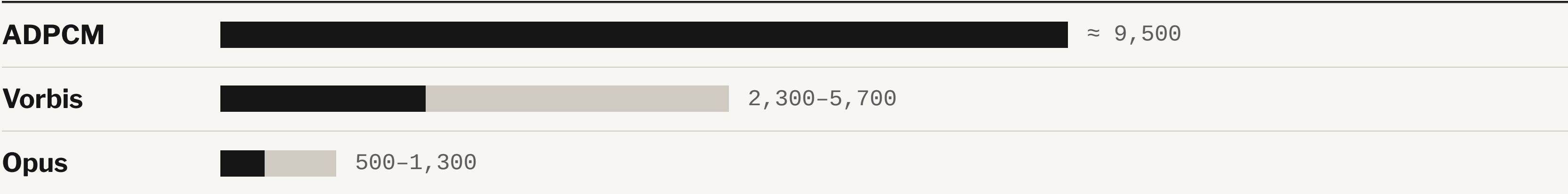
Codec basics: CPU performance

A codec adds a decode step for every playing voice, every time it plays.



PCM skips this only if it already matches the mix format: sample rate, channels, bit depth. Otherwise it gets converted first.

Streams one PC can decode at once (ranges are the quality setting)



Measured by Audiokinetic in Wwise on a 2018 Core i7: [A Guide for Choosing the Right Codec](#)

Codec basics: other gotchas

Size and CPU are not the whole story. Four more things can rule a codec out.

Looping	Some codecs add a sliver of silence at the start and end of a file. A seamless loop clicks or gaps unless your tool trims it. MP3 is the classic case.
Licensing	Some codecs are covered by patents. Vorbis and Opus are royalty-free. AAC is licensed, and MP3 was until 2017.
Proprietary	Some codecs belong to one company: XMA on Xbox, ATRAC9 on PlayStation, Bink Audio in Unreal. Each platform needs its own encode.
Hardware acceleration	Some platforms decode certain codecs on dedicated hardware, which frees CPU for more voices. XMA on Xbox is the usual example.

Codec basics: hardware acceleration

Each platform decodes one specific codec cheaply, usually in dedicated hardware. Only that codec gets the speedup.

Platform	Built-in codec	Notes
Xbox 360, Xbox One	XMA	Up to 320, then 512 voices
Xbox Series X S	Opus	XMA still supported for older content
Nintendo Switch	Opus	A small pool of hardware decoders
PlayStation	ATRAC9	Sony's codec for low decode cost; details are in the SDK docs
Phones	AAC	Hardware decoders on most phones
Everything else	Vorbis, Bink Audio and RAD Audio decode on the CPU	
Run out	Hardware decoders are limited. Past the limit, sounds fail to start	
Seeking	Unreal's "Platform Specific" setting doesn't support seeking	

02

The past

A short history of game audio codecs

The past

Year	Console	Audio memory	Compression	Meanwhile on PC
1990	Super Nintendo	64 KB audio RAM, 8 voices	BRR (ADPCM), about 3.5:1	Sound Blaster: 8-bit mono, ADPCM
1994	PlayStation	512 KB sound RAM, 24 voices	4-bit ADPCM; XA-ADPCM from CD	Music played as CD audio tracks
2000	PlayStation 2	2 MB sound RAM, 48 voices	4-bit ADPCM	Sound Blaster Live!: 64 voices
2005	Xbox 360	Shared RAM, 320 XMA voices	XMA, decoded in hardware	MP3 and Vorbis, decoded on the CPU
2013	PlayStation 4	Shared system RAM	ATRAC9	Wwise and FMOD, decoding on the CPU
2013	Xbox One	Shared RAM, 512 XMA voices	XMA, decoded in hardware	Opus (2012) joins Vorbis
2020	Xbox Series X S	Shared system RAM	Opus, decoded in hardware	Vorbis, Opus, Bink Audio on the CPU

Consoles started with ADPCM in tiny sound RAM, then moved perceptual codecs into hardware decoders. PCs had the memory and CPU to decode in software.



03

The modern toolbox

Opus, Vorbis, AAC, XMA, ATRAC9, and how to choose

Codecs in use today

Codec	Good for	Watch out for	Wwise / Unreal
ADPCM	Short, frequent SFX	Fixed ~4:1; hiss on bright sounds	Both
Vorbis	Music, ambience	Weaker than Opus at low rates	Wwise
Opus	VO, music, low bitrates	More decode CPU than ADPCM	Both
Bink / RAD	Many voices at once; Unreal's default	Proprietary to Epic	Unreal
AAC	Apple targets	Licensing; loop padding	Wwise (Apple)
XMA	Xbox, hardware decode	Xbox only	Wwise; UE Platform Specific
ATRAC9	PlayStation, cheap decode	PlayStation only	Wwise; UE Platform Specific

Support varies by version and platform. Check your build.

Remember Titanfall's 35 GB? Respawn's later Apex Legends plays 80 to 100 voices at once, all in Bink Audio.

ADPCM

Adaptive differential PCM: predict each sample from the ones before it, then store a 4-bit correction.

Size	Fixed 4:1 against 16-bit PCM
Decode	Near-zero CPU
Loops and seeks	Sample-accurate
Sound	Hiss on bright, high-frequency sounds
Use it for	Short, frequent sounds: footsteps, impacts, UI

Still in use

PlayStation emulators still decode the console's ADPCM samples. A fun read on its sound chip:

jsgroth.dev/blog/posts/ps1-spu-part-1



Vorbis

The royalty-free perceptual codec that replaced MP3 in games.

Released	1.0 in 2002, from Xiph.org; royalty-free
Where	Long-time default in Wwise; not offered in Unreal's sound settings
Decode	Cheaper than Opus, far costlier than ADPCM
Quality	Strong at mid to high bitrates; behind Opus at low ones
Use it for	Music, ambience, long streams

Why not MP3?

MP3 never stuck in games. Its encoder adds silence that breaks seamless loops, and it needed a patent licence until 2017.

Vorbis did the same job for free.

Opus

An open, royalty-free codec built for real-time voice that turned out to be good at everything.

Standard	IETF RFC 6716 (2012), royalty-free
How	SILK for speech, CELT for music, or a hybrid
Range	6 to 510 kbps; frames from 2.5 to 60 ms
Strength	Best quality per bit at low bitrates
Hardware	Decoded in hardware on Xbox Series X S
Use it for	Dialogue, music, long streams, voice chat

Against Vorbis

Better quality at low bitrates.

Costs more CPU to decode: 500 to 1,300 streams against Vorbis's 2,300 to 5,700 in Audiokinetic's test.

Default new work to Opus. Keep Vorbis where a pipeline already depends on it.

Bink Audio and RAD Audio

Perceptual codecs from RAD Game Tools, now part of Epic, built for games: small files at close to ADPCM's decode cost.

Bink Audio	Perceptually lossless at 10:1, with decode speeds closer to ADPCM than to MP3 or Vorbis
RAD Audio	Bink Audio with better quality at comparable CPU; 24 to 48 kHz only
In Unreal	Bink Audio is free since UE 4.27 (2021) and the default; RAD Audio was added later
Catch	Proprietary. Free inside Unreal; elsewhere, check licensing with Epic

Performance claims are Epic's: [Unreal Engine blog, June 2021](#) and [Unreal sound compression types](#)

Platform codecs

Codec	Platform	What you get	What you give up
XMA	Xbox	Hardware decode; CPU freed for more voices	Xbox only
ATRAC9	PlayStation	Low latency and low decode cost	PlayStation only
AAC	Apple, Android	Hardware decoders on most phones	Licensing; loop padding

Each platform gets its own encode. Budget build time and QA for it.

**Choose a codec per asset
class, not per project.**

A decision order

Ask in this order. Stop at the first yes.

1	Short, frequent, latency-critical?	ADPCM, or PCM if memory allows
2	Dialogue or voice chat?	Opus, at speech bitrates
3	Long and streamed from disk?	Opus or Vorbis
4	CPU-bound at peak voice count?	Platform codec: XMA, ATRAC9, AAC

On Unreal: start from Bink Audio, the default, for 1, 3 and 4. Voice chat still needs Opus.

On Wwise: Audiokinetic's [A Guide for Choosing the Right Codec](#) has measured CPU and size numbers for every platform.

Then listen on target hardware, at real mix levels, before locking bitrates.

What to measure

Decode CPU	Profile at worst-case voice count, not one sound in isolation.
Memory vs. streaming	Resident assets want cheap decode. Streams want ratio and seeking.
Loops and seeking	Check that sample-accurate loop points survive encoding.
Bitrate per class	Set per asset class, then listen on target speakers and headphones.
Platform rules	Check each platform's format requirements.
Tools and licensing	Confirm encoder support in the pipeline, and any license terms.

04

The frontier

Sound effects, spatial delivery, voice chat and generated voices

Problem: sound effects in memory

QOA: the Quite OK Audio format

3.2

bits per sample

5:1

fixed, against 16-bit PCM

≈ 400

lines of C for the reference codec

- An open, MIT-licensed lossy codec by Dominic Szablewski, released in 2023.
- Works in the time domain like ADPCM: it predicts each sample and stores a 3-bit correction.
- By its author's account: better quality than ADPCM, and 3× faster to decode than Ogg Vorbis.
- The format spec fits on one page.

github.com/phoboslab/qoa

For game audio

- Look for it if you are on Godot (4.3 and later), raylib or a custom engine.
- Try it where you use ADPCM today: short, busy SFX.
- Keep Opus or Vorbis for music and VO. QOA's ratio is fixed.

Problem: delivering a finished spatial mix

Spatial delivery: APAC and IAMF

From Apple

APAC

- Apple Positional Audio Codec, built into Apple's operating systems.
- Carries channel layouts and Ambisonics. Required for Apple Immersive Video.
- Apple platforms only, encoded with Apple's tools.

Open and royalty-free

IAMF

- A container from the Alliance for Open Media. It wraps Opus, AAC, FLAC or PCM.
- Carries channel layouts and Ambisonics, with mixing and loudness instructions.
- Announced for YouTube and Samsung's 2025 TVs, as Eclipsa Audio.

What they are for	A finished mix: cinematics, trailers, immersive video.
What they are not	A runtime format. Your engine still renders the interactive mix.
What to do	Find out where your linear content will play, and deliver what that platform decodes.

Problem: voice chat

Opus 1.6: neural help for voice chat

9 kbps

fullband speech, matching Opus 1.4 at 18

1 second

of backup audio in every packet

≈ 1%

of a laptop CPU core per feature, or less

Hides lost packets	Fills the gap with plausible speech	Deep PLC
Rides out dropouts	Carries up to 1 second of backup audio in each packet	DRED
Clearer speech at low bitrates	Repairs the decoded voice and restores its top end	LACE, NoLACE, BWE

What to look for: all of it is off by default. Ask which of these your engine's or middleware's Opus build turns on.

What to do: test voice chat at low bitrates and with packet loss, not only on a clean office network.

opus-codec.org/demo/opus-1.6

Problem: voice chat

IVAS: immersive voice from the phone standards

13.2–512

kbps bitrate range

32–38 ms

algorithmic delay

3rd order

Ambisonics, plus objects and 7.1+4

- Immersive Voice and Audio Services: the new 3GPP voice codec, in Release 18.
- Extends EVS, the mono codec phones already use for calls, and stays compatible with it.
- Carries stereo, multichannel, Ambisonics, objects, and a parametric spatial format (MASA).
- Renders inside the codec: speakers, or headphones with head tracking.

[nokia.com: IVAS codec](https://nokia.com/IVAS_codec)

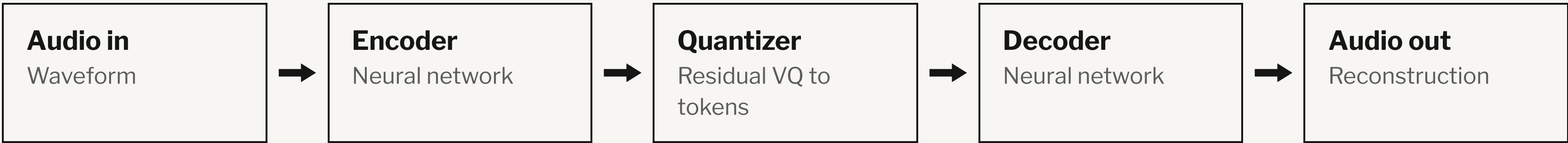
For game audio

- The natural fit is spatial voice chat: placed, head-tracked voices.
- Aimed at calls, conferencing and XR. Games are not a stated target.
- Like APAC, a delivery codec, not a runtime asset format.

Problem: voice chat

Neural codecs

Instead of hand-written rules about hearing, a neural network learns from examples how to shrink sound.



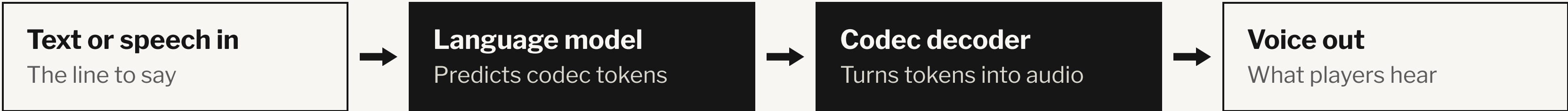
Codec	From	Year	Notes
SoundStream	Google	2021	Streamable and causal; about 3 to 18 kbps
EnCodec	Meta	2022	General audio; about 1.5 to 24 kbps
Lyra	Google	2021–22	Speech only; v2 is built on SoundStream

What to listen for: when they fail, you hear a natural-sounding wrong sound, not hiss or warble.

Problem: generated voices

AI voices are built on codecs

Many voice generators never produce a waveform. A language model predicts codec tokens, and a neural codec's decoder turns them into sound.



Bandwidth	The codec's sample rate caps the result. Ask what rate the voice is generated at.
Latency	Token rate and lookahead set how fast a line starts. Ask for time to first audio.
Where it runs	On device means one model per voice. In the cloud means a network round trip.
Repeatability	The same line can come back different. That matters for QA, lip sync and localization.

The decoder invents plausible detail, so its mistakes sound like a different word or voice, not like noise.

Barriers in games

Cost per voice	One model inference per stream, and a game mixes dozens of voices.
Latency	Many models need lookahead. Streaming variants trade quality for speed.
Training data	Mostly speech and music. Sharp SFX transients are unfamiliar.
Determinism	Float math on GPUs and NPUs can differ across platforms.
Model size	Weights ship with the game and compete for memory.
Tooling	Little middleware support today, and no standard bitstream.

Where neural audio shows up first

Now	ML inside classic codecs. Opus 1.5 and 1.6 added neural packet-loss recovery, speech enhancement and bandwidth extension.
Next	Low-bitrate speech. Research codecs reach 20 ms latency at roughly 20× real time on one CPU.
Speculative	Tokens in place of waveforms. Generative pipelines that store and stream codec tokens, decoded on device NPUs.

Expect it in voice before music or SFX.

Takeaways

- 1 This is an optimization. At the start of a project, your engine's defaults are fine; revisit codecs once you know your budgets.
 - 2 Codec choice is a budget decision: CPU, memory, disk and quality at once.
 - 3 Decide per asset class. ADPCM for short SFX, Opus for VO and streams, platform codecs where hardware helps, and Bink Audio as the starting point on Unreal.
 - 4 On the frontier: QOA for sound effects, APAC and IAMF for spatial delivery, and neural tools reaching voice chat first.
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Thank you

Questions?

Feel free to get in touch



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